

Luis Mesias

Senior Research Engineer | XR, Robotics, Haptics

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SUMMARY

- Senior research engineer building XR, robotics, and rehabilitation systems in Unity/C#, Python, ROS 2, and WebRTC, from research prototypes to field and human-subject studies.
- Co-led immersive teleoperation and haptics architecture with sub-second closed-loop latency and synchronized video/tactile feedback across 2,000+ miles.

SKILLS

Languages: C#, Python, C++, MATLAB

XR/Robotics: Unity, ROS 2, WebRTC, Meta Quest, Varjo, Magic Leap, Ultraleap

Systems: Networking, protocol/serial engineering, .NET Standard 2.0, data analysis, IRB workflows, documentation/training

EXPERIENCE

Case Western Reserve University

May 2026–Present

Senior Research Engineer

- Conduct VR and rehabilitation engineering research using Unity/C#, Varjo, Meta Quest, Python, and data analysis in human-subject research contexts.
- Improved a Meta Quest home-therapy app for veterans with TBI with secure remote data exchange, configurable training schedules, task difficulty, and performance logging.

Human Fusions Institute

Oct 2020–Feb 2026

Research Assistant

- Co-led an operator-side VR/haptics stack for long-distance teleoperation; integrated Unity with ROS 2 and WebRTC to achieve sub-second end-to-end latency.
- Designed a low-profile wearable stimulation interface compatible with optical hand tracking; improved fingertip localization accuracy from 70% to 92% and reduced mislocalized sensations from 27% to 4%.

VA Portland Health Care System

Oct 2025–Dec 2025

VR Software Contractor (Independent)

- Delivered a Unity/C# Magic Leap projection environment with configurable head-/world-centric modes, rotation locking, documentation, and researcher training.

HIGHLIGHTS

- ANA Avatar XPRIZE semifinalist; teleoperation system featured by National Geographic and PBS NewsHour.
- Architected open-source WSS software in the COSMIIC ecosystem with C#, Python, Unity, and ROS 2 APIs for stimulation hardware control.
- Demonstrated hexapod teleoperation with glove control and haptic feedback for tactile sensing of buried targets in sand.

EDUCATION

Ph.D. Electrical Engineering (GPA 3.82), Case Western Reserve University, Cleveland, OH

Aug 2020–May 2026

B.S.E. Electrical Engineering (GPA 3.96), Summa Cum Laude, Case Western Reserve University, Cleveland, OH

Aug 2018–Aug 2020